

3G Network and Security

Module 42

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Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture we will study about 3G network and 3G network security history of 3G network 3G technology is the result of research and development work carried out by International telecommunication Union itu in the early 1980s 3G specifications and standards were developed in 15 years the technical specifications were made available to the public under the name IMT 2000 the communication spectrum between 400 mhz to 3 GHz was allocated for 3G the first pre-commercial 3G network was launched by NTT DoCoMo in Japan in 1998 branded as FOMA F4 the first European pre-commercial network was an UMTS Network on the island of Man by Vodafone

Telecom the first Network to go commercially was by SK Telecom in South Korea on the CDMA based one into EV-DO technology in January 2002 the first commercial United States 3G network was by Verizon mobile networks on CDMA 2001 into EV-DO technology the first pre-commercial demonstration Network in the southern hemisphere was built in Adelaide South Australia by Optus Corporation in February 2002 using UMTS on 2100 mhz Optus launched the first 3G Network in Africa 11th December 2008 India entered the 3G arena with the launch of 3G enabled mobile and data services by government owned Mahanagar Telephone Neam Limited mtnl in Delhi and later in Mumbai mtnl becomes the first 3G mobile service provider in India after mtnl

another state operator BSNL launched 3G services on 22nd Feb 2009 in Chennai and later launched 3G as Nationwide however before the rise of the 3G Evolution a transition was made from GPRS to EDGE or 3G, 1 second edge enhanced GPRS which means GPRS on a Red Bull like high MMS boomed and mobile internet took off into the stratosphere this brings us to 3G technically it's a direct upgrade to EDGE networks but it runs on a wholly different set of frequencies entirely new infrastructure has to be deployed for the successful roll out of 3G what is 3G network 3G is the third generation of Wireless Mobile telecommunications technology this is based on a set of Standards used for mobile devices and mobile telecommunications user services and networks that comply with the

international mobile telecommunications 2,000 that is IMT 2000 specifications by the international telecommunication Union 3G means faster connectivity to the internet via your mobile device the chance to finally have a video call while traveling by train or standing on the beach and enjoying the sunshine sending images videos emails Etc as fast as it can be done via desktop computer the third generation 3G mobile devices and services will transform Wireless Communications into online realtime connectivity 3G Wireless technology allows an individual to have immediate access to location specified services that offer information on demand mobile phones are rapidly becoming the preferred means of personal

communication creating the world's largest consumer electronics Industry 3D finds application in Wireless voice telephony mobile internet access fixed wireless internet access video calls and Mobile TV 3D is the result of development and evolution of GSM standards 3D telecommunication networks support services that provide an information transfer rate of at least 200 kilobit per second later 3G releases often denoted 3.5g and 3.75g also provide mobile broadband excess of several mbits per second to smartphones and mobile mods in laptop computer s this ensures it can be applied to wireless voice telephony mobile internet access fixed wireless internet access video calls and Mobile TV Technologies figure shows the various Technologies related to 3G and their suitability for mobile users

Beyond 3D it is predicted that mobile networks and the wireless communication landscape will be based on various Technologies offering seamless Mobility with cellular network networks fourth generation 4G services will enable Broadband wireless communication at home at the office and on the Move in other words new networks will make the services provided by the web and the internet as well as a variety of other services such as multimedia and entertainment available to mobile users 3D capabilities highspeed data Transmissions symmetrical and asymmetrical data transmission support improved voice quality greater capacity multiple simultaneous Services global roaming across networks improved Security Service flexibility 3G applications wireless internet audio on

demand electronic postcards video conferencing secure mobile Commerce transactions traffic and traveling information location specified 3G Services Services provided by 3D Network in general terms 3D Services comprise of wide bandwidth service such as enhanced communication example messaging email video browsing the web and location specific information like informing users about the availability of stores restaurants gas stations free parking Lots so on near them in addition business users will have a direct access to company networks while traveling or working outside office from marketing point of view identifying and designing these services and setting an appropriate price is said to be the core marketing task related to 3G Mobile internet browsing the web from

mobile the aim to merge cellular networks and the internet meaning that mobile users can have access to all services that the internet provides from messaging to contemporary management research the term mobile internet or internet in Mobile refers to gaining access

to the internet using an handheld wireless device like a mobile phone as shown in the figure or PD as 2G networks have been mainly voice Centric with low data transmission capacity 2.5g and 3G will speed up data transmission speeds however in 3G networks the data transmission speed is dependent upon the number of users accessing the network at the same point of time email Services email service was rated as the number one preferred mobile service by Sweden

in 2001 followed by Banking and use of Encyclopedia email Services can be categorized into web style HDM ml email service and plain text email Services HTML email Services provide more flexibility with the format and appearance while plain text email comprises of a letter style message although email has become a widely acceptable and accessible Communications medium during the recent years there is lack of research about the use of email from mobile devices for many internet users both private and business users email has been the most common way of exchanging information documents and communication although email is cheap easy to access and is asynchronous in nature email Services have suffered from

the receipt of unsolicited messages or even spam the fear of receiving unsolicited messages and spam on mobile devices may slow the adoption of email Services among mobile users second as the first mobile virus news has been reported in media BBC News 2003 mobile users might want to protect their mobile devices by avoiding using email Services messaging services mobile messaging referring to short message service SMS and multimedia message service MMS is expected to be the most utilized mobile service in the future it is estimated that the step by by step consumers as well as business users will upgrade their messaging from sending simple text messages to multimedia messages including pictures and video clips

although 3G networks will make bandwidth hungry services such as sending receiving and downloading video clips possible there are notable cultural differences in adopting multimedia messaging services for example while the European Market has not adopted multimedia messaging to large extent due to various reasons such as pricing policy and relatively complex usage Asians have eagerly adopted MMS on average while a typical user in Europe sends one to two MMS messages per month and Asian takes and sends 20 to 30 MMS messages besides SMS and MMS Services instant messaging I am on the Internet is one of the most popular services especially among youngsters the instant messaging service has the potential to be utilized widely among mobile users as the service allow

users to send content example a text message HTML page a picture a file containing a song a video clip or other file to another user in near real time this instant messaging service is combined with presence service giving the possibility to see if other users are available and reachable the framework in figure shows some of the major Services provided by 3G and underlying factors 3G architecture 3G wireless networks consist of a radio Access Network R and a core Network the core Network consists of a packet switch domain which includes 3G ggsn and ggsn which provide the same functionality that they provide in a GPRS system and a circuit switch domain which includes 3D MSC for switching off voice calls charging for

services and

excess is done through the charging Gateway function cgf which is also part of the core Network AR and functionality is independent from the core Network functionality the excess network provides a core Network Technology independent access for mobile terminals to different types of core networks and Network Services either core Network domain can access any appropriate RN service example it should be possible to access a speech radio access Bearer from the packet switched domain the radio Access Network consist of new network elements known as node B and Radio Network Network controllers rnc's node B is comparable to the base transceiver station in 2G wireless networks RNC replaces the base station controller it provides the radio Resource Management Handover control and

support for the connections to Circuit switched and packet switched domains the interconnection of the network elements in R and between RN and core network is over Lu L and LU interfaces based on ATM as a layer two switching technology data services run from the terminal device over IP which in turn uses ATM as a reliable transport with qos voice is embedded into ATM from the edge of the network node B and is transported over ATM out of the RNC the Lu interface is split into two Parts circuit switched and packet switched the Lu interface is based on ATM with voice traffic embedded on Virtual circuits using A2 technology and IP over ATM for data traffic using a A5 technology these traffic types are switched independently to either 3G gsn

for data or 3D MSC for voice figure shows was the 3G wireless network architecture effect of 4G on 3G network technology nowadays in Indian telecommunication Market 4G network is the hottest topic this excitement has been fueled out by the launch of 4G Reliance Geo considered as the largest investment in Indian history it is important to note that Indian operators invested significantly in 3G network and Spectrum from 2010 2015 slow uptake of 3G in India means a lot of that investment has not yet been fully recovered as 4G becomes the dominant option some operators are considering shutdown of their 3D networks to simplify their architectures however this is happening in countries where typically 3D networks have been in

operation for almost 15 years on the other hand developing countries like Indonesia Thailand Etc continue to focus on 3G even though they have launched 4G networks and are looking for a slow migration to 4G networks India 2 was in a similar stage of evolution until 2015 and even 5 years after launch 3D penetration was less than 10% one key reason was that smartphones and 3G data pricing were both both expensive no wonder we had 300 million internet users of which 180 million had smartphones but only 90 million bought 3G plans network security network security consists of the policies and practices adopted to prevent and monitor unauthorized access misuse modification or denial of a computer network and

network accessible resources network security involves the authorization of access to data in the network which is 17 minutes, 1 second controlled by the network administrator users Choose Or assigned an ID and password or other authenticating information that allows them access to information and programs within their Authority network security covers a variety of computer networks both public and private that are used in everyday jobs conducting transactions and Communications among businesses government agencies and individuals networks can be private such as within a company and others which might be open to public access network security is involved in organizations Enterprises and other types of Institutions it does as its

title explains it secures the network as well as protecting and overseeing operations being done the most common and simple way of prot protecting a network resource is by assigning it a unique name and a corresponding password security involves the following three elements confidentiality to prevent unauthorized disclosure of information to third parties this includes the disclosure of information about resources Integrity to prevent unauthorized modification of resources and maintain the status qu it includes the Integrity of system resources information and personal the alteration of resources like information may be caused by a desire for personal gain or a need for Revenge availability to prevent unauthorized withholding of system resources from those who need

them when they need them communication infrastructure is a vital component of every country it could be targeted in an attack or also with be on only destructive Purpose with the intent to steal sensitive information these days we read nearly all the vulnerabilities that can be exported to undermine the communications back DUIs by governments in the surveillance of mobile devices flaws in authentication protocols that allow you to very easily Pierce our Wi-Fi networks and malware hacking used to control the VIP Communications from our PC 3G offer greater security allowing Mutual authentication between terminals and networks with 3G network growth the mobile will play a similar ordinary role of a computer hackers are able to penetrate mobile devices exactly in the same way they access to our

confidential data on our computer hackers can easily Spire our movements listen to our phone call read our messages access to all our private data this third generation technology brings along with it a vast number of vulnerabilities making it a heaven for hackers and crackers all this while very few consumers were actually aware of the threats to 3G and each operator will need to spend 5 to 10% of its gain in securing 3D Services suffer more security threats and it is necessary to Define new security Solutions at different levels both at the service providers and and the handset manufacturers the 3D networks will make a mobile communication Network and bandwidth equivalent to a computer network which will in turn open the

chances for cyber criminals to carry out attacks at will through the mobile networks from a purely technological perspective 3G networks use the Kasumi block crypto in instead of the older A5 by1 stream Cipher but also Kazumi Cipher has been identified several serious

weaknesses consider also that the increasing of connectivity means a sensible grow of security exposure harder to manage main security problems related to 3G networks imsi is sent in clear text when allocating tmsi to the user the transmission of IMI is not protected IMI is not a security feature a user can be enticed to camp on a false BS once the user camps on the radio channels of a false BS the user is Out Of Reach of the paging signals of SN hijacking outgoing incoming calls in the

networks with the disabled encryption is possible the Intruder poses a man in the middle and drops the user once the call is set up according to etsi ts121 133 specification 3D threats could be classified as unauthorized access to sensitive data violation of confidentiality eavesdropping an intruder intercepts messages without detection masquerading an intruder Hawes an authorized user into believing that they are the legitimate system to obtain confidential information from the user or an intruder Hawes a legitimate system into believing that they are an authorized user to obtain system service or confidential information traffic analysis an intruder observes the time rate length source and destination of

messages to determine a user's location or to learn whether an important business transaction is taking place browsing an intruder searches data storage for sensitive information leakage an intruder obtains sensitive information by exploiting processes with the legitimate access to the data inference an intruder observes a reaction from a system system by sending a query or signal to the system for example an intruder May actively initiate Communications sessions and then obtain access to information through observation of the time rate length sources or destinations of associated messages on the radio interface unauthorized manipulation of sensitive data that is violation of Integrity manipulation of messages

messages may be deliberately modified inserted replayed or deleted by an intruder disturbing or misusing Network Services leading to denial of service or reduced availability intervention an intruder May prevent a authorized user from using a service by jamming the users traffic signaling or Control Data resource exhaustion an intruder May prevent an authorized user from using a service by overloading the service misuse of privileges a user or a serving Network May exploit their privileges to obtain unauthorized services or information abuse of services an intruder May abuse some special service or facility to gain an advantage or to cause disruption to the network repudiation a user or a network denies

actions that have taken place unauthorized access to services Intruders can access Services by masquerading as users or network entities users or network entities can get unauthorized access to Services by misusing their excess rights so friends this was all about 3G network and 3G network security hope to see you in the next lecture till then goodbye thank you